



# S.P.M College, Udantpuri

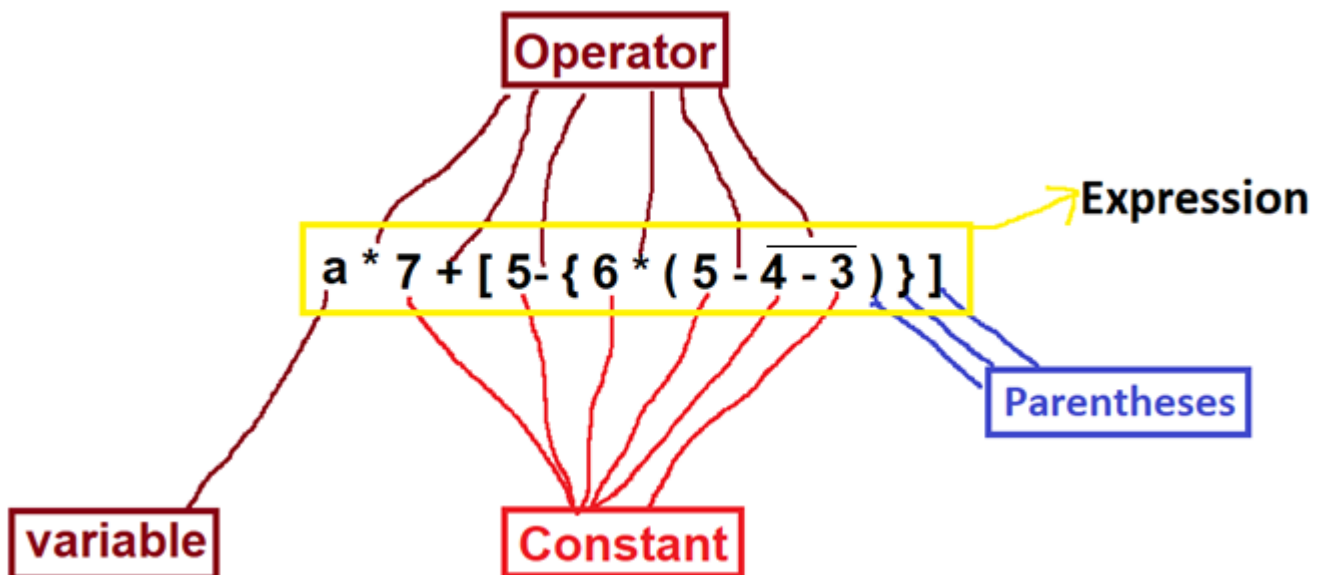
Bachelor Of Computer Application (BCA)

Part – 2 (Paper-3)

– Hira Kumar

## ❖ What is an Expression in C++?

- An expression is a valid arrangement or combination of variables, constants, operators and sometimes parentheses.
- In C++, each expression can be evaluated to compute a value of a given type.
- An expression in mathematics is a finite combination of constant or variable using operators such as addition (+), subtraction (-), multiplication ( $\times$ ), or division ( $\div$ ).



## Types of Expression

- **Numerical Expression**  $\rightarrow$  involves numeric constants and arithmetic operations.  
E.g. (a)  $5 - 2 + 1/2$  (b)  $5 + 2 - (3 \times (-1) + 7)$
- **Algebraic Expression**  $\rightarrow$  variables, constants, and algebraic operators  
E.g. (a)  $3x - 3xy^3$  (monomial expression) (b)  $5xy + 8$  (binomial) (c)  $ax^2 + by + c$  (polynomial)
- **Radical Expression**  $\rightarrow$  use square root, cube root, or fourth root  
E.g. (a)  $3\sqrt{2} - 7$  (b)  $\sqrt{x^2 + 2x - 1}$  (c)  $\sqrt{9x^2} + 1/2$
- **Fractional Expression**  $\rightarrow$  fractions within an algebraic expression  
E.g. (a)  $3/x + 9/20$  (b)  $4x^2/(x - 5)$
- **Trigonometric Expression**  $\rightarrow$  E.g.  $\sin x + \cos x + \tan x$

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- **Logarithmic Expression** → E.g. =  $\log_{10} 100 - \log_{10} 10 + \log_{10} 2$
- **Exponential Expression** → E.g. =  $2^3 - 3^2 + 7^5$

## Expression in Maths

Expression

$$7x + 3 = 4$$

Equation

### ❖ Solve the Following Expressions

1.  $\frac{a*a+a}{a} = 10$
2. Find the coefficient of term y in the expression  $6x^2 + 7y + 3$ .
3. Find the coefficient of the term 7y in the expression  $6x^2 + 7y + 3$ .
4. Find the terms in the expression  $5x + 6y + 3$ .
5. Find the factors in the expression  $7x^3 + 5x^2 + 6x + 2$ .
6. Identify the terms and coefficients in the expression  $3a^2 + 2ab$ .

### Answer

1. 9
2. 7
3. 7
4. 3 (5x, 6y, 3)
5. 4 ( $7x^3$ ,  $5x^2$ , 6x, 2)
6. Terms 2 ( $3a^2$ , 2ab), coefficients 2 (3 & 2)

### ❖ Precedence

- Precedence is the priority of operators, which decides which operator is evaluated first in an expression.
- e.g.  $7 + [5 - \{6 * (5-4-3)\}] = ?$

### ❖ Associativity

- Associativity decides the direction (right to left / left to right ) of evaluation when two or more operators of the same precedence appear in an expression.
- e.g.  $7 + [5 - \{6 * (5-4-3)\}] = ?$

## ❖ Operator in C++

- An operator is a symbol that tells the compiler to perform specific mathematical or logical manipulations.

### Types of Operator

1. Arithmetic Operators (5)
2. Relational Operators (6)
3. Increment / Decrement (2)
4. Logical Operators (3)
5. Bitwise Operators (6)
6. Assignment Operators (11)
7. Ternary or Conditional Operator (1)
8. Special / Miscellaneous Operators (10)

| Operand                                                                             | Operator                                                                                                |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• Any value / Variable is a Operand</li></ul> | <ul style="list-style-type: none"><li>• It is a Symbol / keyword to perform action/operation.</li></ul> |
| <ul style="list-style-type: none"><li>• a,b,c</li></ul>                             | <ul style="list-style-type: none"><li>• + =</li></ul>                                                   |

$$a + b = c$$

Here **a,b,c** Operand **+ =** operator

## 1. Arithmetic Operator

- Arithmetic operators are used to perform arithmetic or mathematical operations on the operands.
- a. **Addition (+)** → Adds two operands  
e.g.  $10 + 10 = 20$
  - b. **Subtraction (-)** → Subtracts second operand from the first.  
e.g.  $10 - 10 = 0$

c. **Multiplication (\*)** → Multiplies two operands

e.g.  $10 * 10 = 100$

d. **Division (/)** → Divides first operand by the second operand.

e.g.  $10 / 10 = 1$

e. **Modulo Operation (%)** → Returns the remainder of an integer division.

e.g.  $10 \% 10 = 0$

| Arithmetic operator | Associativity | Precedence |        |
|---------------------|---------------|------------|--------|
|                     |               | Highest    | lowest |
| $+, -, *, /, \%$    | Left to Right | $* / \%$   | $+ -$  |

## Solve Arithmetic Expression

I.  $10+5*2 = ?$

II.  $20-6/3 = ?$

III.  $6*5/10 = ?$

IV.  $30/5\%4 = ?$

V.  $18\%5+2 = ?$

VI.  $40-20*3/5 = ?$

VII.  $8 + 16/4 * 2 = ?$

VIII.  $25\%7*2 = ?$

IX.  $100/10+6*2 = ?$

X.  $50-10\%4*3 = ?$

## Answer

i → 20, ii → 18, iii → 3, iv → 2, v → 5, vi → 28, vii → 16, viii → 8,

ix → 22, x → 44

# Expression solve

```
1  #include<iostream>
2  #include<conio.h>
3  using namespace std;
4  int main()
5  {
6      int a = 50-10%4*3 ;
7      cout << a;
8      getch();
9
10 }
```

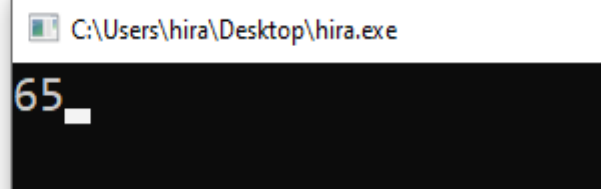


## ❖ Type casting :

- Type casting is the process of converting a variable from one data type to another.
- C++ supports several types of casting:  
E.g. - int → float, char → int (ASCII conversion), float → double....

## Q. Write a C++ Program to the print ASCII Value.

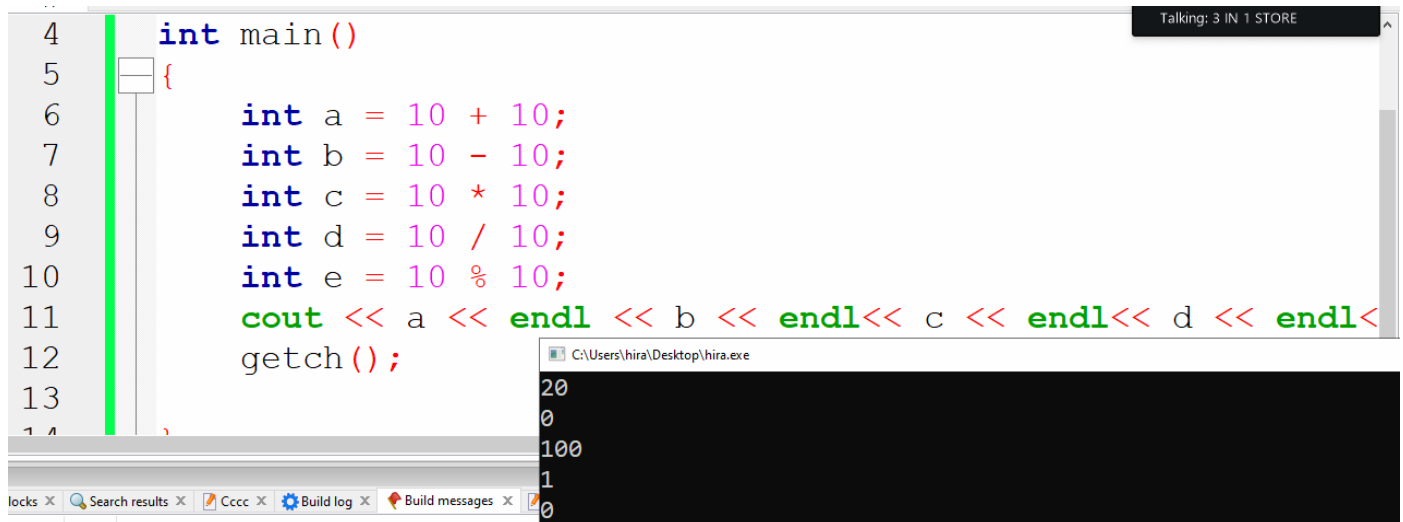
```
1  #include<iostream>
2  #include<conio.h>
3  using namespace std;
4  int main()
5  {
6      char a = 'A';
7      cout << int(a);
8      getch();
9
10 }
```



**Q. Write a C++ program to perform all arithmetic operations on two numbers.**

**OR**

**Write a C++ program to perform addition, subtraction, multiplication, division, and modulus operations on any two numbers.**



```
4 int main()  
5 {  
6     int a = 10 + 10;  
7     int b = 10 - 10;  
8     int c = 10 * 10;  
9     int d = 10 / 10;  
10    int e = 10 % 10;  
11    cout << a << endl << b << endl << c << endl << d << endl << e << endl;  
12    getch();  
13 }
```

Output:

```
20  
0  
100  
1  
0
```

## Home work

**Q1. What will be the output ?**

```
Int main()  
{  
    Int a=15, b= 4;  
    Cout << a+b*2-a%b;  
    Return 0;  
}
```

**Q2. Write a C++ program that takes two number (1<sup>st</sup> number any integers and second number any float) and perform Addition, subtraction, multiplication, integer division, modulo as input and display.**

**Q3. Expression Evaluation, if: int x =20, y= 6;  
Find output of : cout << x/y+x%y\*2;**

#### **Q4. Find the Error**

```
Int main()  
{  
    Int a, b;  
    Cout << "Enter two numbers : ";  
    Cin << a << b;  
    Cout >> a%b;  
    Return 0;  
}
```

**5. Write a C++ program that takes a 3-digit number (example : 147) as input and prints : First digit, Second digit, third digit.**

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